Chapter 4. Sampling Distributions

Binomial, normal, chi-square, t and F distributions

Contents

4.1 The Bernoulli and normal distributions as population distributions **2**
4.1.1 Bernoulli trials . . . 2
4.1.2 The normal distribution . . . 3

4.2 Sampling distributions **5**
4.2.1 Sampling distribution of a statistic . . . 5
4.2.2 Random samples . . . 6

4.3 The binomial distribution **6**

4.4 The distribution of the sample mean **7**
4.4.1 Normal approximation to the binomial . . . 9

4.5 The chi-square, t and F distributions **10**
4.5.1 The chi-square distribution . . . 10
4.5.2 The t distribution . . . 12
4.5.3 The F distribution . . . 13

4.6 Normal quantile plots **14**

4.1 The Bernoulli and normal distributions as population distributions

Motivation. An estimator is judged by its distribution, driven by the population model: Bernoulli or normal.

4.1.1 Bernoulli trials

An experiment with two outcomes (pass/fail, cured/not), labelled success s and failure f, is a **Bernoulli trial**.

Definition 4-1 (Bernoulli trial)

(a) A Bernoulli trial has exactly two outcomes, success s and failure f, so its sample space is the two-point set $S = \{s, f\}$.
(b) The success probability is $p = P(\{s\})$ and the failure probability is $q = P(\{f\}) = 1 - p$, where $0 \le p \le 1$.

Definition 4-2 (Bernoulli random variable)

$X(s) = 1$, $X(f) = 0$ on $S = \{s, f\}$ is a **Bernoulli random variable**, with the **Bernoulli distribution** $P(X = 1) = p$, $P(X = 0) = 1 - p$.

The Bernoulli distribution models a population whose characteristic is dichotomous (e.g. patients "cured"/"not cured" coded 1/0).

Example 4-1 (Mean and variance of the Bernoulli distribution)

Find the mean and variance of the Bernoulli distribution.
Solution. $E(X) = 1 \cdot p + 0 \cdot (1 - p) = p$ and $E(X^2) = 1^2 \cdot p + 0^2 \cdot (1 - p) = p$. Hence

$$\mathrm{Var}(X) = E(X^2) - \{E(X)\}^2 = p - p^2 = p(1 - p).$$

Bernoulli trials are **independent** if the result of one trial does not affect the others, and **repeated** if the success probability is the same in every trial.

Illustration 4-1: sampling with replacement

Two of 10 cards (3 winning) are drawn with replacement; X_1, X_2 are the Bernoulli variables of the draws. Then

$$P(X_2 = 1) = P(X_2 = 1 \mid X_1 = 1) = P(X_2 = 1 \mid X_1 = 0) = \tfrac{3}{10},$$

and $P(X_1 = 1) = \frac{3}{10}$, so X_1, X_2 are independent and identically distributed; likewise for n draws with replacement.

Illustration 4-2: sampling without replacement

Draw the two cards of Illustration 4-1 *without* replacement. Then $P(X_1 = 1) = \frac{3}{10}$, $P(X_2 = 1 \mid X_1 = 1) = \frac{2}{9}$, $P(X_2 = 1 \mid X_1 = 0) = \frac{3}{9}$, and by the law of total probability

$$P(X_2 = 1) = \tfrac{2}{9} \cdot \tfrac{3}{10} + \tfrac{3}{9} \cdot \tfrac{7}{10} = \tfrac{3}{10}.$$

So each draw wins with probability $\frac{3}{10}$, but $P(X_2 = 1) \neq P(X_2 = 1 \mid X_1 = 1)$: the trials are not independent.

Illustration 4-3: general finite deck

With N cards of which M are winning, sampling two without replacement gives

$$P(X_1 = 1) = P(X_2 = 1) = \frac{M}{N}, \qquad P(X_2 = 1 \mid X_1 = 1) = \frac{M-1}{N-1},$$

so again the draws are identically distributed but dependent. If N (and M) is large relative to the sample, $P(X_2 = 1) \approx P(X_2 = 1 \mid X_1 = 1)$ and the draws are nearly independent.

Sampling with vs. without replacement

From a finite population, sampling without replacement is practically indistinguishable from sampling with replacement when the population size N is large compared with the sample size.

4.1.2 The normal distribution

The normal distribution is the leading model for an infinite population whose characteristic is continuous. Its density curve

(a) is symmetric about a single point, the population mean μ;

(b) is highest at the centre μ;

(c) decreases as one moves away from μ;

(d) has inflection points at $\mu \pm \sigma$.

Definition 4-3 (Normal density with mean μ and standard deviation σ)

$$p(x) = \frac{1}{\sqrt{2\pi}\,\sigma} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}, \qquad -\infty < x < \infty.$$

We write $X \sim N(\mu, \sigma^2)$.

μ shifts the curve; smaller σ makes it narrower and taller. An observation from a normal population has $X \sim N(\mu, \sigma^2)$.

Theorem 4-1 (Properties of the normal distribution 1)

If $X \sim N(\mu, \sigma^2)$, then for any constants $a \neq 0$ and b,

$$aX + b \sim N(a\mu + b,\ a^2\sigma^2).$$

Proof. [Supplement] We use moment generating functions (mgf), $M_X(t) = E(e^{tX})$; an mgf finite near 0 determines the distribution.

Step 1. For $X \sim N(\mu, \sigma^2)$, substituting $x = \mu + \sigma z$ and completing the square $tz\sigma - \frac{z^2}{2} = -\frac{1}{2}(z - \sigma t)^2 + \frac{\sigma^2 t^2}{2}$,

$$M_X(t) = e^{\mu t} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{\sigma t z - z^2/2}\, dz = e^{\mu t + \sigma^2 t^2/2} \int_{-\infty}^{\infty} \frac{e^{-(z-\sigma t)^2/2}}{\sqrt{2\pi}}\, dz = e^{\mu t + \sigma^2 t^2/2}.$$

Step 2. Hence $M_{aX+b}(t) = e^{bt}M_X(at) = e^{(a\mu+b)t+a^2\sigma^2t^2/2}$, which is the mgf of $N(a\mu+b, a^2\sigma^2)$. By uniqueness of mgfs the claim follows. ■

(If $a = 0$ then $aX + b = b$ is constant, a "normal distribution with variance 0".)

In particular, taking $a = 1/\sigma$, $b = -\mu/\sigma$,

$$X \sim N(\mu, \sigma^2) \implies Z = \frac{X-\mu}{\sigma} \sim N(0,1). \tag{4.1}$$

$N(0,1)$ is the **standard normal distribution**; a standard normal variable is usually written Z. Appendix C tabulates $P(Z \le z)$.

Definition (Probabilities from the standard normal table)

(a) $P(a \le Z \le b) = P(Z \le b) - P(Z \le a)$.
(b) $P(Z \ge z) = 1 - P(Z < z)$.

Example 4-2 (Two tails)

Find $P(Z < -1.9 \text{ or } Z > 2.1)$.
Solution. The events are disjoint, so the probability is $P(Z < -1.9) + P(Z > 2.1) = 0.0287 + (1 - 0.9821) = 0.0287 + 0.0179 = 0.0466$.

Example 4-3 (A quantile)

Find z with $P(Z > z) = 0.025$.
Solution. $P(Z \le z) = 1 - 0.025 = 0.975$, so the table gives $z = 1.96$.

Definition (Upper quantile z_α)

For $0 < \alpha < 1$, z_α is the number with $P(Z > z_\alpha) = \alpha$, $Z \sim N(0,1)$; it is the $100(1-\alpha)$th percentile (quantile) of $N(0,1)$. Frequently used values:

$$z_{0.005} = 2.58, \qquad z_{0.025} = 1.96, \qquad z_{0.05} = 1.645.$$

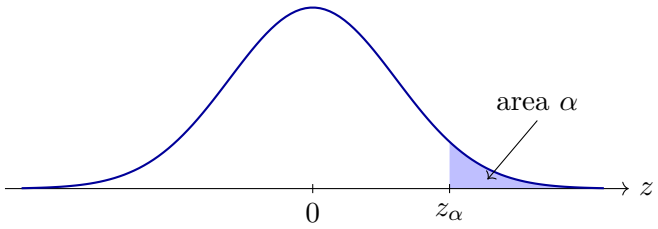


Figure 1: The upper quantile z_α of $N(0,1)$.

If $X \sim N(\mu, \sigma^2)$, standardizing with (4.1) gives

$$P(X \le b) = P\Big(Z \le \frac{b-\mu}{\sigma}\Big),$$
$$P(a \le X \le b) = P\Big(\frac{a-\mu}{\sigma} \le Z \le \frac{b-\mu}{\sigma}\Big).$$

Example 4-4 (Proportion of A grades)

Grades in a statistics class are approximately $N(60, 10^2)$. Students scoring 75 or more get an A. Approximate the proportion of A grades.
Solution. $P(X \ge 75) = P(\frac{X-60}{10} \ge \frac{75-60}{10}) \approx P(Z \ge 1.5) = 0.0668$.

Theorem 4-2 (Properties of the normal distribution 2)

If $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$ are independent, then for any constants a_1, a_2,

$$a_1X_1 + a_2X_2 \sim N(a_1\mu_1 + a_2\mu_2,\ a_1^2\sigma_1^2 + a_2^2\sigma_2^2).$$

Proof. [Supplement] By independence the mgf of a sum is the product of the mgfs. Using Step 1 of the proof of Theorem 4-1,

$$\begin{aligned} M_{a_1X_1+a_2X_2}(t) &= M_{X_1}(a_1t)\, M_{X_2}(a_2t) = e^{a_1\mu_1 t + a_1^2\sigma_1^2t^2/2}\, e^{a_2\mu_2 t + a_2^2\sigma_2^2t^2/2} \\ &= \exp\Big\{(a_1\mu_1 + a_2\mu_2)t + \tfrac{1}{2}(a_1^2\sigma_1^2 + a_2^2\sigma_2^2)t^2\Big\}, \end{aligned}$$

the mgf of the stated normal distribution. By induction the same holds for any finite number of independent normal variables: $\sum a_iX_i \sim N(\sum a_i\mu_i, \sum a_i^2\sigma_i^2)$. ■

Example 4-5 (Calories burned)

Calories burned running are approximately $N(200, 15^2)$ and cycling $N(80, 5^2)$, independently. Find the probability that the daily total is at least 300 kcal.
Solution. The total S is approximately normal with mean $200 + 80 = 280$ and variance $15^2 + 5^2 = 250$. So

$$P(S \ge 300) = P\Big(\frac{S - 280}{\sqrt{250}} \ge \frac{300 - 280}{\sqrt{250}}\Big) \approx P(Z \ge 1.27) = 1 - 0.898 = 0.102.$$

4.2 Sampling distributions

Motivation. An estimation method is judged over *all* possible samples, by the distribution of the statistic.

4.2.1 Sampling distribution of a statistic

A survey estimates p by the sample proportion; accuracy is the probability, over random samples, that the error is at most a given amount.

Illustration: all samples of size 3

Population {yes, no, yes, no, yes} (true $p = 0.6$), sample of size 3 without replacement.

Sample	Probability	Sample proportion	Error
{yes, yes, yes}	1/10	1	0.4
{yes, yes, no}	6/10	2/3	0.067
{yes, no, no}	3/10	1/3	−0.267

So $P(|\text{error}| \le 0.1) = 0.6$ and $P(|\text{error}| \le 0.3) = 0.9$.

Definition (Statistic and sampling distribution)

A **statistic** is a random variable computed from the sample. The probability distribution of a statistic is its **sampling distribution**; it quantifies how reliable an estimate is.

Example 4-6 (Sample mean from a small population)

From the finite population $\{2, 2, 4, 5\}$ draw a sample of size 2 without replacement. Find the distribution of the sample mean.
Solution. The six equally likely samples (probability $\frac{1}{6}$ each) and their means are

Sample	$\{2,2\}$	$\{2,4\}$	$\{2,5\}$	$\{2,4\}$	$\{2,5\}$	$\{4,5\}$
Mean	2	3	3.5	3	3.5	4.5

Hence $P(\bar{X} = 2) = \frac{1}{6}$, $P(\bar{X} = 3) = \frac{1}{3}$, $P(\bar{X} = 3.5) = \frac{1}{3}$, $P(\bar{X} = 4.5) = \frac{1}{6}$.

The sampling distribution depends on both the population distribution and the sampling method (with or without replacement).

4.2.2 Random samples

A simple random sample from a finite population is a **random sample** $X_1, \ldots, X_n$ (categorical answers coded 1/0). With replacement, $X_1, \ldots, X_n$

(a) each have the population distribution, and

(b) are mutually independent.

When the population is large, a random sample without replacement nearly has properties (a) and (b); this motivates the definition for infinite populations.

Definition 4-4 (Random sample)

(a) Finite population: a sample drawn by simple random sampling without replacement.
(b) Infinite population: random variables $X_1, \ldots, X_n$ such that (i) each X_i has the population distribution and (ii) $X_1, \ldots, X_n$ are independent.

Large finite populations are treated as infinite; from Chapter 5 on, all inference assumes a random sample from an infinite population.

4.3 The binomial distribution

Motivation. The number of successes in a random sample has the binomial distribution.

For a 0/1 population with proportion p, a random sample is n independent Bernoulli(p) trials, with number of successes

$$X = X_1 + \cdots + X_n.$$

Its distribution is the **binomial distribution** $B(n, p)$; $B(1, p)$ is the Bernoulli distribution.

Definition 4-5 (Binomial distribution)

With n trials and success probability p, the probability mass function is

$$p(x) = \binom{n}{x} p^x (1-p)^{n-x}, \qquad x = 0, 1, \ldots, n.$$

$\binom{n}{x}$ is the coefficient of $a^{n-x}b^x$ in $(a+b)^n$ (Pascal's triangle). $B(n, 0.5)$ is symmetric; $p < 0.5$ gives **right skew**, $p > 0.5$ **left skew**.

Example 4-7 (Guessing on a multiple-choice test)

A test has 20 five-choice questions and every answer is guessed at random. Find the probability of
(a) no correct answer;
(b) 4 to 6 correct answers;
(c) at least 2 correct answers.
Solution. The number correct is $X \sim B(20, 0.2)$.
(a) $P(X = 0) = \binom{20}{0} 0.2^0\, 0.8^{20} = 0.012$.
(b) $P(4 \le X \le 6) = \sum_{x=4}^{6} \binom{20}{x} 0.2^x\, 0.8^{20-x} = 0.502$.
(c) $P(X \ge 2) = 1 - P(X = 0) - P(X = 1) = 1 - 0.8^{20} - 20(0.2)(0.8^{19}) = 0.931$.

Example 4-8 (On-time deliveries)

A courier delivers on time 95% of the time. In a random sample of 20 deliveries, find the probability that the sample on-time rate is at most 90%.
Solution. The number of late deliveries X is approximately $B(20, 0.05)$. An on-time rate $\le 90\%$ means a late rate $X/20 \ge 0.1$, so

$$P\Big(\frac{X}{20} \ge 0.1\Big) = P(X \ge 2) = 1 - P(X = 0) - P(X = 1) = 0.264.$$

Theorem 4-3 (Mean and variance of $B(n, p)$)

If $X \sim B(n, p)$, then
(a) $E(X) = np$;
(b) $\mathrm{Var}(X) = np(1-p)$.

Proof. Write $X = X_1 + \cdots + X_n$ with X_i independent Bernoulli(p), so $E(X_i) = p$ and $\mathrm{Var}(X_i) = p(1-p)$ (Example 4-1).

(a) By linearity of expectation, $E(X) = \sum_i E(X_i) = np$.

(b) For independent variables the covariances vanish, so $\mathrm{Var}(X) = \sum_i \mathrm{Var}(X_i) = np(1-p)$. ■

4.4 The distribution of the sample mean

Motivation. The sample mean is exactly normal for normal populations and approximately normal for large n (CLT).

For a 0/1 population, $\bar{X}$ is the sample proportion (§4.3). Let $X_1, \ldots, X_n$ be a random sample from

an infinite population with mean μ and variance σ^2, and

$$\bar{X} = \frac{1}{n}(X_1 + \cdots + X_n).$$

Theorem 4-4 (Distribution of the sample mean 1)

For a random sample of size n from a population with mean μ and variance σ^2,
(a) $E(\bar{X}) = \mu$;
(b) $\mathrm{Var}(\bar{X}) = \sigma^2/n$;
(c) $\mathrm{sd}(\bar{X}) = \sigma/\sqrt{n}$.

Proof.

(a) Since $E(X_i) = \mu$, $E(\bar{X}) = \frac{1}{n}\{E(X_1) + \cdots + E(X_n)\} = \frac{1}{n} \cdot n\mu = \mu$.

(b) Since the X_i are independent with $\mathrm{Var}(X_i) = \sigma^2$, $\mathrm{Var}(\bar{X}) = \frac{1}{n^2}\{\mathrm{Var}(X_1) + \cdots + \mathrm{Var}(X_n)\} = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}$.

(c) Take square roots in (b). ■

Finite population

For a sample of size n drawn without replacement from a finite population of size N with mean μ and variance σ^2,

$$E(\bar{X}) = \mu, \qquad \mathrm{Var}(\bar{X}) = \frac{N-n}{N-1} \cdot \frac{\sigma^2}{n}.$$

Thus $\bar{X}$ is centred at μ and concentrates around μ as n grows.

Theorem 4-5 (Distribution of the sample mean 2)

If the population is $N(\mu, \sigma^2)$, then $\bar{X} \sim N(\mu, \sigma^2/n)$.

Proof. By the extension of Theorem 4-2 to n variables, $\bar{X} = \sum_{i=1}^{n} \frac{1}{n} X_i$ is normal with mean $\sum \frac{1}{n}\mu = \mu$ and variance $\sum \frac{1}{n^2}\sigma^2 = \sigma^2/n$. ■

Theorem 4-6 (Distribution of the sample mean 3: central limit theorem)

For a random sample of size n from any population with mean μ and variance σ^2, when n is large $\bar{X}$ is approximately $N(\mu, \sigma^2/n)$:

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \dot{\sim} N(0, 1).$$

Proof. [Supplement] We prove the precise statement $P(\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \le z) \to P(Z \le z)$ for every z, under the extra assumption that the mgf $M(t) = E(e^{t(X_1-\mu)/\sigma})$ is finite for $|t| < h$. We use the *continuity theorem*: if $M_{W_n}(t) \to M_W(t)$ for all $|t| < h$, then the cdf of W_n converges to that of W at every continuity point.

Step 1. Let $Y_i = (X_i - \mu)/\sigma$, so the Y_i are i.i.d. with $E(Y_i) = 0$, $E(Y_i^2) = 1$, and

$$W_n = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} Y_i, \qquad M_{W_n}(t) = \left\{ M\left(\frac{t}{\sqrt{n}}\right) \right\}^n.$$

Step 2. By Taylor's theorem, $M(s) = 1 + M'(0)s + \frac{1}{2}M''(0)s^2 + r(s)$ with $r(s)/s^2 \to 0$ as $s \to 0$. Here $M'(0) = E(Y_1) = 0$ and $M''(0) = E(Y_1^2) = 1$, so

$$M\Big(\frac{t}{\sqrt{n}}\Big) = 1 + \frac{t^2}{2n} + r\Big(\frac{t}{\sqrt{n}}\Big), \qquad n\,r\Big(\frac{t}{\sqrt{n}}\Big) = t^2 \cdot \frac{r(t/\sqrt{n})}{(t/\sqrt{n})^2} \to 0.$$

Step 3. Hence $M_{W_n}(t) = \big(1 + \frac{a_n}{n}\big)^n$ with $a_n = \frac{t^2}{2} + n\,r(t/\sqrt{n}) \to \frac{t^2}{2}$, and since $(1 + a_n/n)^n \to e^a$ whenever $a_n \to a$, $M_{W_n}(t) \to e^{t^2/2}$, the mgf of $N(0,1)$.

Step 4. By the continuity theorem, and since the cdf of $N(0,1)$ is continuous, $P(W_n \le z) \to P(Z \le z)$ for every z. (Without the mgf assumption the same argument works with characteristic functions $E(e^{itY})$.) ■

The striking point is that the conclusion holds whatever the shape of the population; this is why the normal distribution is central to inference.

Illustration 4-4: the CLT in action

From the uniform population on $0, 1, \ldots, 9$ ($\mu = 4.5$, $\sigma^2 = 8.25$), 100 samples of size 5 give a roughly bell-shaped histogram of $\bar{X}$ centred near 4.5 (relative frequencies 0.29, 0.27 in $[4,5)$, $[5,6)$, tapering to 0.03, 0.07), even for $n = 5$.

How large n must be depends on how far the population is from normal. For a normal population $\bar{X}$ is exactly normal; otherwise $n > 30$ usually gives a satisfactory normal approximation.

Example 4-9 (Calories of a new drink)

A new drink has mean 168 kcal and standard deviation 6 kcal. For a random sample of 100 drinks, find $P(167 \le \bar{X} \le 169)$.
Solution. With $n = 100$ the CLT applies; $\bar{X}$ has mean 168 and standard deviation $6/\sqrt{100} = 0.6$. So

$$\begin{aligned} P(167 \le \bar{X} \le 169) &\approx P\Big(\frac{167 - 168}{0.6} \le Z \le \frac{169 - 168}{0.6}\Big) \\ &= P(-1.67 \le Z \le 1.67) = 0.9525 - 0.0475 = 0.9050. \end{aligned}$$

4.4.1 Normal approximation to the binomial

For a random sample from $B(1,p)$, $X = \sum X_i \sim B(n,p)$, with mean p, variance $p(1-p)$ and $\bar{X} = \hat{p} = X/n$. The CLT gives:

Definition (Normal approximation to the binomial)

If $X \sim B(n,p)$ and n is large (rule of thumb: $np \ge 5$ and $n(1-p) \ge 5$), then X is approximately $N(np, np(1-p))$:

$$\frac{X - np}{\sqrt{np(1-p)}} = \frac{\hat{p} - p}{\sqrt{p(1-p)/n}} \dot{\sim} N(0,1).$$

For example $B(15, 0.6)$ is already well approximated by $N(9, 3.6)$ since $np = 9 > 5$.

Example 4-10 (Normal approximation)

Let $X \sim B(100, \frac{1}{5})$. Approximate
(a) $P(X \le 21)$;
(b) $P(15 \le X \le 19)$.

Solution. $np = 20$ and $\sqrt{np(1-p)} = \sqrt{100 \cdot \frac{1}{5} \cdot \frac{4}{5}} = 4$.
(a) $P(X \le 21) \approx P(Z \le \frac{21-20}{4}) = P(Z \le 0.25) = 0.5987$.
(b) $P(15 \le X \le 19) \approx P(-1.25 \le Z \le -0.25) = 0.4013 - 0.1056 = 0.2957$.

Continuity correction

A binomial probability is a sum of bars of width 1 centred at the integers. For $X \sim B(15, 0.6)$, $P(7 \le X \le 10)$ is better approximated by the normal area from 6.5 to 10.5:

$$P(7 \le X \le 10) = P(6.5 \le X \le 10.5) \approx P\Big(\frac{6.5-9}{\sqrt{3.6}} \le Z \le \frac{10.5-9}{\sqrt{3.6}}\Big)$$
$$= P(-1.318 \le Z \le 0.791) = 0.692,$$

very close to the exact value 0.688. This adjustment is the **continuity correction**.

4.5 The chi-square, t and F distributions

Motivation. Statistics built from the sample variance follow chi-square, t and F distributions.

For a random sample $X_1, \dots, X_n$ from $N(\mu, \sigma^2)$ the **sample variance** is

$$S^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X})^2.$$

4.5.1 The chi-square distribution

Definition 4-6 (Chi-square distribution)

If $Z_1, \dots, Z_k$ is a random sample from $N(0,1)$, the distribution of $Z_1^2 + \cdots + Z_k^2$ is the **chi-square distribution with k degrees of freedom**:

$$Z_1^2 + \cdots + Z_k^2 \sim \chi^2(k).$$

The chi-square density has a long right tail. The upper quantile $\chi^2_\alpha(k)$ satisfies $P(V \ge \chi^2_\alpha(k)) = \alpha$; e.g. $\chi^2_{0.995}(4) = 0.21$, $\chi^2_{0.975}(4) = 0.48$, $\chi^2_{0.05}(4) = 9.49$.

Theorem 4-7 (Additivity of the chi-square distribution)

If $V_1 \sim \chi^2(k_1)$ and $V_2 \sim \chi^2(k_2)$ are independent, then $V_1 + V_2 \sim \chi^2(k_1 + k_2)$.

Proof. [Supplement]

Step 1. For $Z \sim N(0,1)$ and $t < \frac{1}{2}$, substituting $u = z\sqrt{1-2t}$,

$$E(e^{tZ^2}) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-(1-2t)z^2/2}\, dz = (1-2t)^{-1/2}.$$

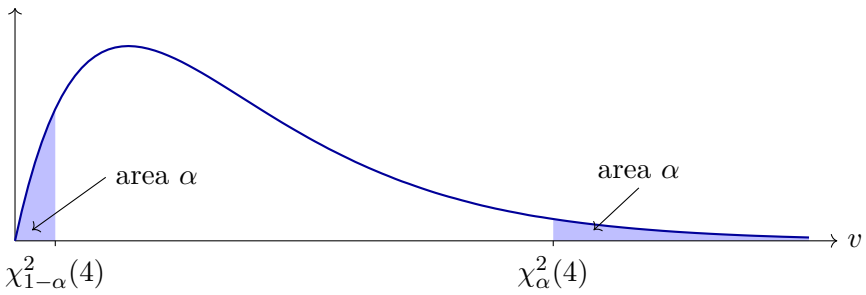


Figure 2: The $\chi^2(4)$ density $ve^{-v/2}/4$ with $\alpha = 0.05$: $\chi^2_{0.95}(4) = 0.71$, $\chi^2_{0.05}(4) = 9.49$.

Step 2. Hence, by independence, $V \sim \chi^2(k)$ has mgf $M_V(t) = (1-2t)^{-k/2}$, $t < \frac{1}{2}$; conversely any variable with this mgf has the $\chi^2(k)$ distribution (uniqueness of mgfs).

Step 3. For independent V_1, V_2, $M_{V_1+V_2}(t) = (1-2t)^{-k_1/2}(1-2t)^{-k_2/2} = (1-2t)^{-(k_1+k_2)/2}$, so $V_1 + V_2 \sim \chi^2(k_1+k_2)$. ■

Theorem 4-8 (Distribution of the sample variance from a normal population)

For a random sample $X_1, \dots, X_n$ from $N(\mu, \sigma^2)$ with sample variance S^2,

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1).$$

Moreover $\bar{X}$ and S^2 are independent.

Proof. The book's argument: since $(X_i - \mu)/\sigma \sim N(0,1)$ independently, $\sum_{i=1}^n \left(\frac{X_i-\mu}{\sigma}\right)^2 \sim \chi^2(n)$. Adding and subtracting $\bar{X}$ (the cross term vanishes because $\sum(X_i - \bar{X}) = 0$),

$$\sum_{i=1}^{n} \left(\frac{X_i-\mu}{\sigma}\right)^2 = \frac{(n-1)S^2}{\sigma^2} + \left(\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}\right)^2,$$

and the last term is $\chi^2(1)$. "Subtracting" degrees of freedom gives $\chi^2(n-1)$. We make this rigorous.

[Supplement]

Step 1. Let $Z_i = (X_i - \mu)/\sigma$, independent $N(0,1)$, so $\mathbf{Z} = (Z_1, \dots, Z_n)^\top$ has joint density $(2\pi)^{-n/2} e^{-\|\mathbf{z}\|^2/2}$. Let A be an $n \times n$ orthogonal matrix whose first row is $(1/\sqrt{n}, \dots, 1/\sqrt{n})$ (complete this unit vector to an orthonormal basis), and $\mathbf{W} = A\mathbf{Z}$.

Step 2. The map $\mathbf{z} \mapsto A\mathbf{z}$ has Jacobian $|\det A| = 1$ and preserves $\|\mathbf{z}\|$, so $\mathbf{W}$ has the same density: $W_1, \dots, W_n$ are independent $N(0,1)$.

Step 3. $W_1 = \sqrt{n}\,\bar{Z}$ and $\sum_{i=1}^n W_i^2 = \sum_{i=1}^n Z_i^2$, hence

$$\sum_{i=2}^{n} W_i^2 = \sum_{i=1}^{n} Z_i^2 - n\bar{Z}^2 = \sum_{i=1}^{n} (Z_i - \bar{Z})^2 = \frac{(n-1)S^2}{\sigma^2}.$$

Step 4. By Definition 4-6 this is $\chi^2(n-1)$. It is a function of $W_2, \dots, W_n$, while $\bar{X} = \mu + \sigma W_1/\sqrt{n}$ is a function of W_1; so $\bar{X}$ and S^2 are independent. ■

Next take independent random samples $X_1, \dots, X_{n_1}$ from $N(\mu_1, \sigma^2)$ and $Y_1, \dots, Y_{n_2}$ from $N(\mu_2, \sigma^2)$ (common variance), with

$$S_1^2 = \frac{1}{n_1 - 1}\sum_{i=1}^{n_1}(X_i - \bar{X})^2, \qquad S_2^2 = \frac{1}{n_2 - 1}\sum_{i=1}^{n_2}(Y_i - \bar{Y})^2.$$

Definition (Pooled sample variance)

$$S_p^2 = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2},$$

the weighted average of S_1^2, S_2^2 with weights $\frac{n_j-1}{(n_1-1)+(n_2-1)}$; it estimates the common variance σ^2.

Theorem 4-9 (Sample variances from two normal populations with equal variance)

Under the above assumptions,

$$\frac{(n_1+n_2-2)S_p^2}{\sigma^2} \sim \chi^2(n_1+n_2-2).$$

Proof. By Theorem 4-8, $(n_1-1)S_1^2/\sigma^2 \sim \chi^2(n_1-1)$ and $(n_2-1)S_2^2/\sigma^2 \sim \chi^2(n_2-1)$, and they are independent since they come from independent samples. Their sum is $(n_1+n_2-2)S_p^2/\sigma^2$, which is $\chi^2(n_1+n_2-2)$ by Theorem 4-7. ■

4.5.2 The t distribution

When σ is unknown we replace it by the sample standard deviation S and obtain the **studentized** sample mean $\frac{\bar{X}-\mu}{S/\sqrt{n}}$.

Definition 4-7 (t distribution)

If $Z \sim N(0,1)$ and $V \sim \chi^2(k)$ are independent, the distribution of

$$T = \frac{Z}{\sqrt{V/k}}$$

is the t **distribution with** k **degrees of freedom**, $T \sim t(k)$.

Like $N(0,1)$, $t(k)$ is symmetric about 0, but it has heavier tails; it is used especially for small samples. The upper quantile $t_\alpha(k)$ satisfies $P(T \geq t_\alpha(k)) = \alpha$; e.g. $t_{0.1}(4) = 1.533$, $t_{0.05}(4) = 2.132$, $t_{0.01}(4) = 3.747$.

Theorem 4-10 (Distribution of the studentized sample mean)

For a random sample $X_1, \ldots, X_n$ from $N(\mu, \sigma^2)$,

$$\frac{\bar{X}-\mu}{S/\sqrt{n}} \sim t(n-1).$$

Proof. Dividing numerator and denominator by $\sigma/\sqrt{n}$,

$$\frac{\bar{X}-\mu}{S/\sqrt{n}} = \frac{(\bar{X}-\mu)/(\sigma/\sqrt{n})}{\sqrt{\dfrac{(n-1)S^2}{\sigma^2}\Big/(n-1)}}.$$

The numerator is $N(0,1)$ (Theorem 4-5), the root contains $\chi^2(n-1)/(n-1)$ (Theorem 4-8), independent of it (Theorem 4-8; the book omits this proof); so the ratio is $t(n-1)$ (Definition 4-7). ■

For two normal populations with equal variance, a short computation gives

$$\frac{(\bar{X}-\bar{Y})-(\mu_1-\mu_2)}{S_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}} = \frac{\frac{(\bar{X}-\bar{Y})-(\mu_1-\mu_2)}{\sigma\sqrt{1/n_1+1/n_2}}}{\sqrt{\frac{(n_1+n_2-2)S_p^2}{\sigma^2}\Big/(n_1+n_2-2)}}.$$

Theorem 4-11 (t distribution for two normal populations with equal variance)

Under the assumptions of Theorem 4-9,

$$\frac{(\bar{X}-\bar{Y})-(\mu_1-\mu_2)}{S_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}} \sim t(n_1+n_2-2).$$

Proof.

Step 1. Numerator. $\bar{X} \sim N(\mu_1, \sigma^2/n_1)$ and $\bar{Y} \sim N(\mu_2, \sigma^2/n_2)$ are independent, so by Theorem 4-2 $\bar{X}-\bar{Y} \sim N(\mu_1-\mu_2, \sigma^2(\frac{1}{n_1}+\frac{1}{n_2}))$, and the standardized numerator in the display is $N(0,1)$.

Step 2. Denominator. By Theorem 4-9, $(n_1+n_2-2)S_p^2/\sigma^2 \sim \chi^2(n_1+n_2-2)$.

Step 3. Independence. By Theorem 4-8, $\bar{X}$ is independent of S_1^2 and $\bar{Y}$ of S_2^2; as the two samples are independent, $(\bar{X}, \bar{Y})$ is independent of (S_1^2, S_2^2), hence of S_p^2.

Step 4. Definition 4-7 gives $t(n_1+n_2-2)$. ■

4.5.3 The F distribution

Definition 4-8 (F distribution)

If $V_1 \sim \chi^2(k_1)$ and $V_2 \sim \chi^2(k_2)$ are independent, the distribution of

$$F = \frac{V_1/k_1}{V_2/k_2}$$

is the F **distribution with** (k_1, k_2) **degrees of freedom**, $F \sim F(k_1, k_2)$.

The F density is right-skewed on $(0, \infty)$. Its upper quantile $F_\alpha(k_1, k_2)$ satisfies $P(F \geq F_\alpha(k_1, k_2)) = \alpha$; e.g. $F_{0.10}(3,4) = 4.19$, $F_{0.05}(3,4) = 6.59$.

Theorem 4-12 (Property of the F distribution)

If $F \sim F(k_1, k_2)$, then $1/F \sim F(k_2, k_1)$. Consequently

$$F_{1-\alpha}(k_2, k_1) = \frac{1}{F_\alpha(k_1, k_2)}.$$

Proof.

Step 1. Write $F = \frac{V_1/k_1}{V_2/k_2}$ as in Definition 4-8. Then $1/F = \frac{V_2/k_2}{V_1/k_1}$ is a ratio of the same form with the roles of (V_1, k_1) and (V_2, k_2) exchanged, so $1/F \sim F(k_2, k_1)$.

Step 2. Let $F' = 1/F \sim F(k_2, k_1)$. Since $F > 0$, $\alpha = P(F \geq F_\alpha(k_1, k_2)) = P(F' \leq 1/F_\alpha(k_1, k_2))$, so $P(F' \geq 1/F_\alpha(k_1, k_2)) = 1-\alpha$ (F' is continuous). By definition of the upper quantile, $F_{1-\alpha}(k_2, k_1) = 1/F_\alpha(k_1, k_2)$. ■

For example, $F_{0.95}(4,3) = 1/F_{0.05}(3,4) = 1/6.59 = 0.15$.

For independent samples from $N(\mu_1, \sigma_1^2)$ and $N(\mu_2, \sigma_2^2)$, $(n_1-1)S_1^2/\sigma_1^2 \sim \chi^2(n_1-1)$ and $(n_2-1)S_2^2/\sigma_2^2 \sim \chi^2(n_2-1)$ independently.

Theorem 4-13 (Ratio of sample variances from two normal populations)

Under these assumptions,

$$\frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F(n_1-1, n_2-1).$$

Proof. Put $V_j = (n_j-1)S_j^2/\sigma_j^2 \sim \chi^2(n_j-1)$ (Theorem 4-8); V_1, V_2 are independent. Then $\frac{V_1/(n_1-1)}{V_2/(n_2-1)} = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2}$, which is $F(n_1-1, n_2-1)$ by Definition 4-8. ■

Theorem 4-14 (Relation between the F and t distributions)

If $T \sim t(k)$, then $T^2 \sim F(1, k)$.

Proof. Write $T = Z/\sqrt{V/k}$ with $Z \sim N(0,1)$, $V \sim \chi^2(k)$ independent. Then $T^2 = \frac{Z^2/1}{V/k}$, where $Z^2 \sim \chi^2(1)$ (Definition 4-6) and Z^2 is independent of V. By Definition 4-8, $T^2 \sim F(1, k)$. ■

4.6 Normal quantile plots

Motivation. A normal quantile (Q–Q) plot checks the normality assumed later.

A q-quantile splits the area $q : (1-q)$. A **normal quantile plot** (Q–Q plot) plots standard normal quantiles against the data quantiles.

Definition (Normal quantile plot)

Order the data $x_{(1)} < x_{(2)} < \cdots < x_{(n)}$. The value $x_{(i)}$ is roughly the $\frac{i}{n+1}$-quantile of the data, whose standard normal counterpart is $z_{1-i/(n+1)}$. Plot the points

$$\left(z_{1-i/(n+1)},\, x_{(i)}\right), \qquad i = 1, \ldots, n.$$

If the points lie close to a straight line, the normality assumption is reasonable. Software often uses $\frac{i-3/8}{n+1/4}$ instead of $\frac{i}{n+1}$.

E.g. for $n = 99$ from $N(2, 3^2)$ the population quantiles $3z_{0.99}+2, \ldots, 3z_{0.01}+2$ are linear in the normal quantiles, so the points $(z_{0.99}, x_{(1)}), \ldots, (z_{0.01}, x_{(99)})$ lie near a line of slope 3, intercept 2.

Reading a Q–Q plot

Systematic departures from the line indicate a non-normal population:
(a) light tails: the extreme points bend toward the centre (an S-shape flatter than the line at both ends);
(b) heavy tails: the extreme points fall below the line on the left and above it on the right;
(c) long right tail: the points curve upward on the right;
(d) long left tail: the points curve downward on the left.